Problem 10 | Problems T5_MechanicalProperties.pdf #10 | Yield · unloading · resilience

Copper Wire: Yield Strength, Permanent Elongation after Unloading and Resilience

Original statement · Problems T5_MechanicalProperties.pdf, p. 4

10. Plastic deformation in a copper wire of 200 cm in length and a diameter of 1 mm begins when it is loaded with 15.3 kg. If the elastic modulus for copper is 110 GPa.

- *a. Calculate its yield strength.*
- *b. Considering that when it is loaded with 29.6 kg the strain is 0.011, calculate the final length of the wire once it has been unloaded.*
- *c. Calculate the resilience modulus.*

Official Solution: a. $\sigma_y = 190.9$ **MPa; b.** $l = 201.85$ **cm; c.** $U_r = 0.165$ **MPa**

HYPOTHESES & PARAMETERS

Hypotheses

1. **Uniaxial, uniform stress** in the wire: $\sigma = F/A_0$ [Session 11 Slide 9]. The weight of a mass m is $F = m\,g$ with $g = 9.81$ m/s².
2. **Yield strength.** "Plastic deformation begins when it is loaded with 15.3 kg" gives the yield strength, $\sigma_y = F_y/A_0$ [Session 11 Slide 28].
3. **Total strain under load.** The strain 0.011 of item b is the total strain (elastic plus plastic) measured while the load of 29.6 kg is applied, relative to the initial length: $\varepsilon = \Delta l/l_0$ [Session 11 Slide 9].
4. **Unloading is elastic.** When the load is removed the elastic strain is recovered along a line of slope E, parallel to the elastic line, starting from the stress that the wire carried; what remains is the plastic strain [Session 11 Slides 21 and 32].
5. **Resilience** of the elastic range: $U_r = \sigma_y^2/(2E)$ [Session 11 Slide 35].

Data

Quantity	Symbol	Value	In SI
Length	l_0	200 cm	2.000 m = 2000 mm
Diameter	d_0	1 mm	1×10^{-3} m
Young's modulus	E	110 GPa	110 000 N/mm²
Mass at which plastic deformation begins	m_1	15.3 kg	$F_1 = 15.3 \times 9.81 = 150.09$ N
Mass of item b	m_2	29.6 kg	$F_2 = 29.6 \times 9.81 = 290.38$ N

Quantity	Symbol	Value	In SI
Total strain under m_2	ε	0.011	0.011 (dimensionless)

Unknowns: σ_y (a), the final length after unloading (b), U_r (c).

MODEL & STRATEGY

One direction (the axis of the wire); no change of basis.

Strategy. (a) is a stress: weight over area. (c) follows from (a) and E. Item (b) is the delicate one, and it is solved by following the strain along the load path:

1. The wire is loaded up to $\sigma_2 = F_2/A_0$, above the yield strength, and the total strain there is 0.011.
2. It is unloaded along a straight line of slope E: the strain recovered is σ_2/E (the stress that it carried, not the yield strength).
3. The strain that remains is permanent, and it gives the final length.

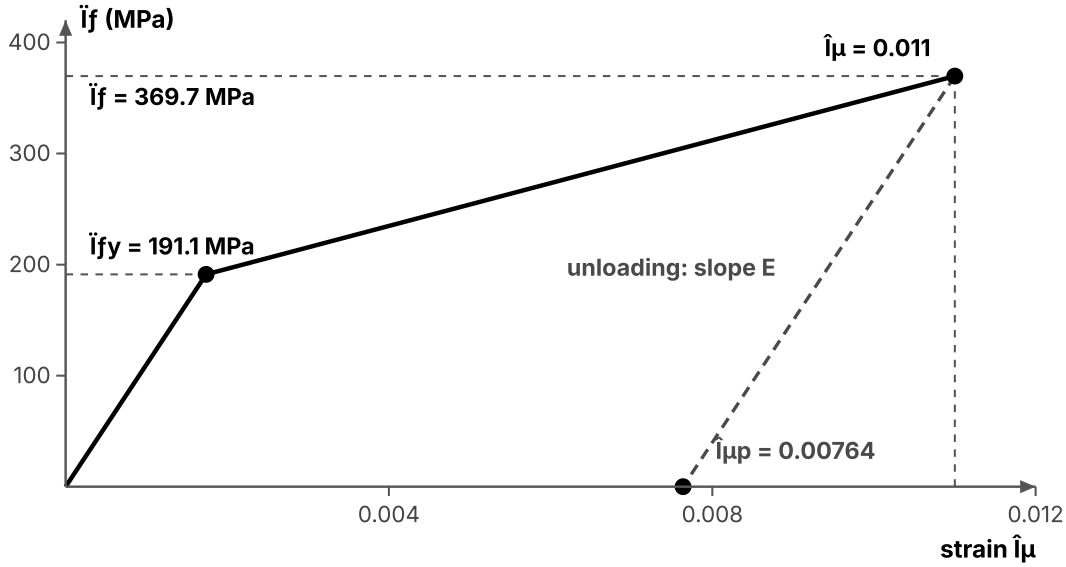


Figure 9. Problem 10: loading beyond the yield point and unloading along a line of slope E; the strain that remains is permanent (the shape between the yield point and the loaded point is not given by the statement).

STEP-BY-STEP CALCULATION

1. Area of the wire

$$A_0 = \frac{\pi d_0^2}{4} = \frac{\pi\,(1\ \text{mm})^2}{4} = 0.7854\ \text{mm}^2$$

a) Yield strength

The force at which the plastic deformation begins is

$$F_1 = m_1\,g = 15.3\ \text{kg} \times 9.81\ \text{m/s}^2 = 150.09\ \text{N}$$

$$\sigma_y = \frac{F_1}{A_0} = \frac{150.09\ \text{N}}{0.7854\ \text{mm}^2} = \mathbf{191.1\ MPa}$$

b) Length after unloading

The stress under the load of 29.6 kg is

$$F_2 = 29.6 \times 9.81 = 290.38\ \text{N} \qquad \sigma_2 = \frac{290.38}{0.7854} = 369.7\ \text{MPa}$$

which is above the yield strength (93 % higher), so the wire is in the plastic zone and the total strain, $\varepsilon = 0.011$, is the sum of an elastic part and a plastic part, $\varepsilon = \varepsilon_e + \varepsilon_p$. On unloading, the elastic part is recovered along the line of slope E that starts at σ_2:

$$\varepsilon_e = \frac{\sigma_2}{E} = \frac{369.7\ \text{N/mm}^2}{110000\ \text{N/mm}^2} = 3.3611 \times 10^{-3}$$

The strain that remains is

$$\varepsilon_p = \varepsilon - \varepsilon_e = 0.011 - 3.3611 \times 10^{-3} = 0.007639$$

and the final length, once unloaded, is

$$l_f = l_0\,(1 + \varepsilon_p) = 200\ \text{cm} \times (1 + 0.007639) = \mathbf{201.53\ cm}$$

(that is, the wire is left 1.53 cm longer than at the beginning).

c) Modulus of resilience

$$U_r = \frac{\sigma_y^2}{2E} = \frac{(191.1\ \text{MPa})^2}{2 \times 110000\ \text{MPa}} = \mathbf{0.166\ MPa}$$

($1\ \text{MPa} = 1\ \text{MJ/m}^3$.)

RESULT & CHECKS

Checks

- **Units.** kg·m/s² = N; N/mm² = MPa; the strains have no units; l_f in the unit of l_0 (cm).
- **Order.** $\varepsilon_e = 3.3611 \times 10^{-3} < \varepsilon = 0.011$, and $\varepsilon_p > 0$: the wire does not return to 200 cm. If the load were removed from the yield point, the permanent strain would be zero; here the wire has gone beyond it.
- **Elastic strain at yield.** The strain of the yield point, $\sigma_y/E = 0.001737$, is smaller than the elastic strain recovered from σ_2 (3.3611×10^{-3}): the unloading starts from a higher stress, and the recovery is larger.
- **Limits.** If $\sigma_2 = \sigma_y$ the wire would be exactly at the yield point and ε_p would be (almost) zero; the larger the stress reached, the larger the elastic recovery and the permanent strain.

Interpretation

The copper wire yields at about 191.1 MPa, and after being loaded to 369.7 MPa it keeps a permanent elongation of 1.53 cm in 200 cm. A load that goes beyond the yield strength leaves a trace: the length after unloading depends on how far the stress went.

VALUE OF g

The key is consistent with $g = 9.8$ m/s², which gives $\sigma_y = 190.9$ MPa instead of 191.1 MPa (0.1 % difference) and $U_r = 0.1657$ MPa. The result here uses $g = 9.81$ m/s² throughout the sheet.

Check against the official solution

Item	Official	Computed	Comment
a. σ_y	190.9 MPa	191.1 MPa	agrees (the key uses $g = 9.8$ m/s²)
b. final length	201.85 cm	201.53 cm	**does not agree**, see below
c. U_r	0.165 MPa	0.166 MPa	agrees

DISCREPANCY WITH THE OFFICIAL SOLUTION: LENGTH AFTER UNLOADING

The key, 201.85 cm, is obtained by recovering only the elastic strain of the yield point, $\sigma_y/E = 1.7373 \times 10^{-3}$: $l = 200\,(1 + 0.011 - 1.7373 \times 10^{-3}) = 201.85$ cm. That is the unloading of a wire that had been stressed to σ_y. But the wire of item b carried $\sigma_2 = 369.7$ MPa, and on unloading it recovers $\sigma_2/E = 3.3611 \times 10^{-3}$. The correct permanent strain is therefore 0.007639 and the length 201.53 cm. The key therefore recovers less elastic strain than the wire really does. The computed value is kept.